Meeting Minutes 11

Project Name: Financial and Operational Analysis of U.S. Hospitals Using Hospital Provider Cost Report Data (2011–2022)

Meeting Minutes 11A (In-Class – Final Model Review & Deployment Strategy)

(Production Pipeline Validation & Application Design Guidance)

Date and Time

31 Mar 2026, 8:00 AM – 9:00 AM

Location

St. Clair College, One Riverside Drive, Room 1004

Attendees

• Professor Manjari Maheshwari
• Manjunath Muthineni (0878795)
• Rajesh Thota (0873782)
• Neha (0873038)
• Abhishekh Choudhary (0873460)
• Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

• Present final model pipeline
• Validate model performance and reproducibility
• Discuss deployment strategy and user interface design

Discussion Points

• Manjunath presented the final training pipeline, which included only the two selected models (Random Forest and ExtraTrees). The pipeline demonstrated consistent preprocessing, feature selection, model training, and saving of serialized model files for deployment.

• Rajesh confirmed that the final model performance metrics were consistent with previously observed cross-validation results, indicating that the models generalized well and were not overfitted to specific data splits.

• Neha explained the structure of the production pipeline and highlighted the importance of maintaining consistency between training and deployment environments to prevent errors during prediction.

• Abhishekh emphasized the importance of feature name alignment between the training dataset and the deployed application, noting that mismatches could lead to runtime errors or incorrect predictions.

• Krishna demonstrated successful loading of trained model files and verified that predictions could be generated correctly using the pipeline.

• Professor Maheshwari provided critical feedback on the usability of the application. She noted that having too many input fields would create friction for users and reduce the practical usability of the system. She advised simplifying the interface by reducing the number of required inputs and deriving additional features programmatically.

• The professor recommended implementing either a simple Streamlit interface for rapid prototyping and ease of use or a Flask application for more customizable deployment. The emphasis was placed on creating a clean, intuitive, and user-friendly interface suitable for non-technical stakeholders.

Decisions Made

• Final model pipeline approved without modification.
• No further model retraining required.
• UI simplification strategy confirmed using derived features and default values.
• Streamlit selected as the preferred framework for development, with Flask as an alternative option.

Action Items

• Implement simplified user interface with reduced input fields.
• Integrate trained models into the application backend.
• Ensure full pipeline consistency and error handling.

Next Meeting Date and Time

Internal development and testing session

Minutes Prepared By

Neha

Meeting Minutes 11B (Group Meeting – Application Development & Testing)

(Streamlit Implementation & End-to-End Validation)

Date and Time

31 Mar 2026, 7:00 PM – 9:30 PM

Location

Online (Microsoft Teams)

Attendees

• Manjunath Muthineni (0878795)
• Rajesh Thota (0873782)
• Neha (0873038)
• Abhishekh Choudhary (0873460)
• Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

• Develop application interface
• Implement feature derivation logic
• Test end-to-end prediction pipeline

Discussion Points

• Manjunath developed the application using Streamlit, structuring it into multiple pages including prediction input, batch processing, model information, and project overview. The design focused on usability and clarity for end users.

• Rajesh implemented the derive_features() function, which calculates additional model inputs based on a reduced set of user-provided inputs. This significantly simplified the interface while preserving model performance.

• Neha resolved a column mismatch issue that arose due to inconsistencies between training and deployment datasets. The issue was fixed by regenerating model files with standardized feature naming conventions.

• Abhishekh validated the prediction outputs using sample test data and confirmed that the results were within expected ranges for both target variables.

• Krishna tested the batch upload functionality, verifying that users could upload datasets, generate predictions for multiple records, and download the results successfully.

Decisions Made

• Final application structure confirmed and validated.
• User interface simplified to improve usability and reduce input complexity.
• End-to-end pipeline confirmed to be functioning correctly.

Action Items

• Perform final UI improvements and styling.
• Complete project documentation and user guide.
• Prepare final submission package.

Next Meeting Date and Time

Final review and submission preparation

Minutes Prepared By

Abhishekh Choudhary